

Summarization and Simplification of Medical Articles using Natural Language Processing

Problem Statement: The paper addresses the challenge of improving *health literacy* by easing access to complex healthcare information by summarising medical texts and simplifying them lexically by translating specific medical terminology to laymen's terms.

Algorithms Used: Used 4 pretrained transformer-based models - *ALBERT*, *DistilBERT*, *SciBERT*, *GPT-2* for extracting summaries from articles and another pretrained model - *en_ner_bionlp13cg_md* for Named Entity Recognition of the medical terms. Used *Lemmatization* for each medical entity and finally *Web Scraping* to generate a simplified meaning.

Datasets: Dataset of 400 medical articles taken from the *Cochrane Database of Systematic Reviews (CDSR)*.

Model Training and Testing: In generated summary, the medical entities present in the summary are highlighted in green and their respective meaning is displayed to the user when he hovers over the highlighted word. *ROUGE i.e. Recall-Oriented Understudy for Gisting Evaluation* metric has been used to evaluate the summaries produced by the models.

Results: The compression rate of the models is 40%. Below figure shows a comparative analysis of the ROUGE scores obtained by the four chosen pretrained models.

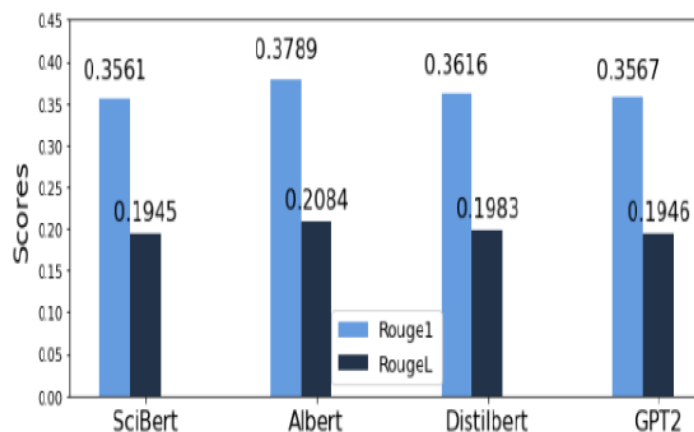


Fig. 5. Comparative analysis of the ROUGE-1 and ROUGE-L scores

Conclusions: Authors employ an extractive text summarization approach using ALBERT. Lexical simplification of the summary is done using Named Entity Recognition along with simplifying the complex words through web scraping.

Open Questions: The paper suggests future work on improving the paragraph summaries and identification of non-medical complex words and medical abbreviations.

Relevance to Our Team: The evaluations into the various transformer based models can guide our team in selecting appropriate models for text summarization which can help clinical researchers to quickly navigate and review documents.

Extracting Health Evidence Information from Biomedical Literature using LLMs

Problem Statement: This article addresses the challenge of bridging the gap between research findings and real-world healthcare applications by exploring *LLMs* to enhance medical practice efficiency and advance *evidence based medicine*.

Algorithms Used:

- *INSTRUCTOR* and *OpenAI* for embeddings
- *FAISS* for storing vector embeddings in a knowledge base
- *PICO* Extraction Algorithm(User-Defined)
- *Stuff* for document processing
- *Streamlit* Framework to design an interface for user engagement
- *LangChain* for summarising complex PDFs
- *RAG*-based pipeline
- *RAGAs* for evaluation of retriever and LLMs
- OpenAI LLMs: *GPT-3.5 turbo*, *GPT-4*, *GPT-4 turbo*, and *GPT-4o* to generate responses

Datasets:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3952010/>

<https://link.springer.com/article/10.1208/s12249-021-02058-y>

[https://www.thelancet.com/journals/landia/article/PIIS2213-8587\(22\)00387-4/abstract](https://www.thelancet.com/journals/landia/article/PIIS2213-8587(22)00387-4/abstract)

<https://onlinelibrary.wiley.com/doi/full/10.1002/pst.2020>

<https://www.sciencedirect.com/science/article/pii/S0753332223014488>

Model Training and Testing: Overall, 12 PICO queries were generated from the documents. These queries were then passed to each LLM to extract evidence using the PICO framework and answer the queries with the extracted evidence.

Results: The average performance of each LLM is detailed in below figure:

Model	Embeddings	AVG Context Recall	AVG Context Precision	AVG Faithfulness	AVG Answer relevancy
GPT 3.5 turbo	Instructor	75.0%	95.5%	93.7%	82.4%
GPT 3.5 turbo	OpenAI	75.0%	99.7%	85.00%	82.3%
GPT 4	Instructor	85.0%	95.0%	96.2%	85.0%
GPT 4	OpenAI	75.0%	100%	96.7%	88.3%
GPT 4 turbo	Instructor	71.0%	94.0%	95.2%	87.6%
GPT 4 turbo	OpenAI	90.0%	99.7%	89.6%	87.8%
GPT 4o	Instructor	71.0%	94.2%	76.3%	85.2%
GPT 4o	OpenAI	75.0%	99.7%	72.67%	87.2%

Conclusions:

- LLMs like GPT-4 and GPT-4 Turbo can effectively handle PICO-based queries in systematic medical reviews.
- The choice of embeddings significantly influences performance.

Open Questions: The predefined limits on context retrieval can result in incomplete information. Future work should aim to increase the context size to take full advantage of models like GPT-4 Turbo, even though this may come with higher costs which can be addressed by their fine-tuning.

Relevance to Our Team: This article is very relevant for our project as it caters to all the aspects of our objective and presents some challenges which we can work on.

Completing A Systematic Review in Hours instead of Months with Interactive AI Agents

Problem Statement: This article addresses accuracy, relevance and quality issues with existing automatic summarization methods due to lack of domain expertise and intervention. It introduces a human-centered interactive AI agent *InsightAgent* powered by LLMs for SR workflow. This agent also provides visualizations of the corpus and agent trajectories, allowing users to monitor the actions of the agent and provide real-time feedback.

Algorithms Used:

- *RSS map* for visualizing corpus structure
- *K-means clustering* for corpus partitioning into semantically distinct clusters
- LLMs: *GPT-4o*, *Llama 3.3 70B*
- *VA* based interface

Datasets: Literature corpus for 15 SRs from PubMed with size ranging from 72 to 7,356 articles

Model Training and Testing: Insight agent is evaluated with both the LLMs in autonomous and user interaction modes. Also, comparison is done with existing agents like BM25, ChatCite and AutoSurvey. Evaluation is done by 9 medical experts on the basis of:

- **Record Screening** - Precision, Recall and F1 scores
- **SR Quality** – Rubric based method
- **User Experience** – Feedback Questionnaire and Interviews

Results: The average performance of each agent is detailed in below figures:

4.2.1 Record Screening

Quality of the Generated Summaries

System	Recall (%)	Precision (%)	F1 %
BM25 (Top-100)	54.30	16.50	25.3
ChatCite	–	–	–
AutoSurvey (Top-100)	70.30	20.43	31.6
<i>InsightAgent_{auto}</i> (Llama 3.3)	66.40	62.40	64.3
<i>InsightAgent</i> (Llama 3.3)	<u>87.90</u>	80.10	<u>83.8</u>
<i>InsightAgent_{auto}</i> (GPT-4o)	71.10	51.90	60.0
<i>InsightAgent</i> (GPT-4o)	98.50	<u>79.80</u>	88.2

System	Score
ChatCite (GPT-4)	47.1
AutoSurvey (GPT-4o)	54.0
<i>InsightAgent_{auto}</i> (Llama 3.3)	60.9
<i>InsightAgent</i> (Llama 3.3)	<u>70.2</u>
<i>InsightAgent_{auto}</i> (GPT-4o)	62.4
<i>InsightAgent</i> (GPT-4o)	79.7

Question	<i>InsightAgent_{auto}</i> (Llama 3.3)	<i>InsightAgent</i> (Llama 3.3)	<i>InsightAgent_{auto}</i> (GPT-4o)	<i>InsightAgent</i> (GPT-4o)
System was easy to use.	2.0/5	<u>3.9/5</u>	2.1/5	4.0/5
Confidence in recommendations.	2.6/5	<u>4.2/5</u>	2.8/5	4.5/5
Visualizations aided understanding.	2.9/5	4.2/5	2.9/5	4.2/5
Ability to guide or correct agents.	2.7/5	<u>3.9/5</u>	2.9/5	4.6/5
Overall satisfaction.	3.0/5	<u>4.0/5</u>	3.2/5	4.3/5

Conclusions: A single domain expert can finish a high quality SR in only 1.5 hours, rather than months, and reach 79.7% of human-written quality.

Open Questions:

- It is a small scale study.
- Considering LLM cost and context length constraint, only title and abstract of each article is used in this study.
- Extraction of numerical statistics or weight evidence is missing.

Relevance to Our Team: This article focuses on rapid SRs reducing review time from days/weeks to hours which is one of the main objectives of our project.